AXI4-Stream ⇄ PCIe-6-style Link: RTL to GDSII on Sky130

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SARCathon 2026 Semiconductor Challenge · **Stage 2 (Grand Finale)**: complete, verify, optimize, implement · Design `axis_pcie_link_mux2_top` (SystemVerilog, 17 files, 2,373 RTL lines) · Open-source flow: Yosys, OpenROAD, OpenSTA, KLayout, Magic, Verilator, Icarus · PDK: SkyWater Sky130 HD · **Every number in this report is copied from a tool report in the submitted ZIP; nothing is estimated.**

Scope and claims. A **PCIe-6.0-inspired** link datapath following the challenge block diagram; **not PCIe-compliant** (no TLP/DLLP, FLIT, LTSSM, PIPE or SerDes; FEC is a bypass placeholder). Data ports are AXI4-Stream (as drawn in the block diagram); full AXI4 memory-mapped master/slave is not implemented — the memory-mapped port is an AXI4-Lite CSR slave.

338 / 338	**16 / 16**	**+0.026 ns**	**0 / 0**	**GDSII**
RTL test runs PASS (Icarus + Verilator/SVA)	gate-level sim PASS on the **routed** netlist	post-route setup slack TT, 100 MHz (hold +0.096)	router DRC / antenna violations after repair	generated (KLayout + Magic); DRC sign-off **open**

1. Status at submission — what is complete and what is not

Stage 2 requirement (PDF)	What was done	Status
Complete RTL integration	All blocks of the selected scope integrated in one top: 2-source arbiter → link layer (FIU, CRC, credits) → 4-lane Gray-PAM4 → RX path → AXIS; AXI4-Lite CSR	Done (scope: no N×M crossbar, FEC stub)
Functional verification, read/write & data transfer, reset & robustness	3 self-checking benches, byte-exact scoreboard, fault injection, SVA, 110/110 coverage bins, 41/41 mutants killed, formal proof of the arbiter	Done
Application-level workloads	Mixed-size bursty traffic (4–3000 B), 64 B vs 4096 B competition, 1200-packet order test; no test is labelled per use case	Partly
RTL / implementation optimization	Two complete PnR iterations; port buffering + repair margins removed all TT slew/cap violations (§7)	Done (measured)
Synthesis + analysis	Yosys on Sky130 HD, formal mapping-equivalence check, gate-level simulation	Done
Physical design	OpenROAD floorplan → place → CTS → route → antenna repair → RC extraction, IR drop, 3-corner STA	Done
Final GDSII	GDSII streamed out (KLayout and Magic) — `physical_design/outputs/`	Generated
Physical sign-off (DRC / LVS)	KLayout DRC: 1,522 violations (met3 minimum area at stacked vias, root cause found). LVS: run started, not completed in time	Open (§8)

2. Architecture

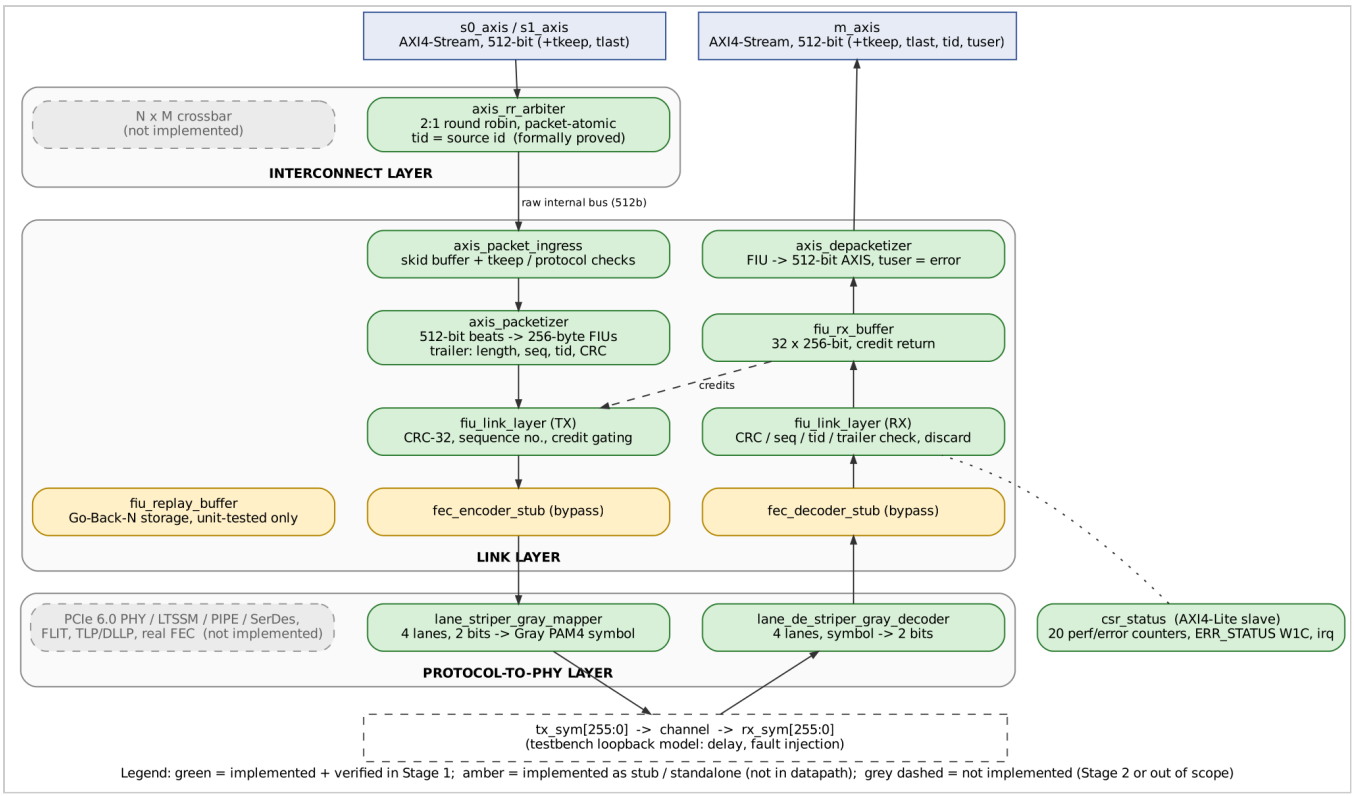


Fig. 1 — Implemented architecture mapped onto the challenge's reference layers (green: implemented and verified; amber: stub / standalone; grey: not implemented).

Datapath. Two 512-bit AXI4-Stream sources are merged by a packet-atomic round-robin arbiter (source → `tid`). The link layer segments packets into **256-byte Flow Information Units** (7 payload words + 1 trailer with length, sop/eop, sequence, magic, `tid`, CRC-32), gates transmission on receiver credits, and stripes each 256-bit word over **4 lanes as Gray-coded PAM4 symbols** (2 bits/symbol). The receiver inverts the mapping, checks CRC/sequence/framing, and commits, rolls back or aborts per FIU in a 32-entry buffer so corrupted data is never delivered as good data (a packet that loses a later FIU is delivered truncated with `tuser=1`).

3. RTL design and interfaces

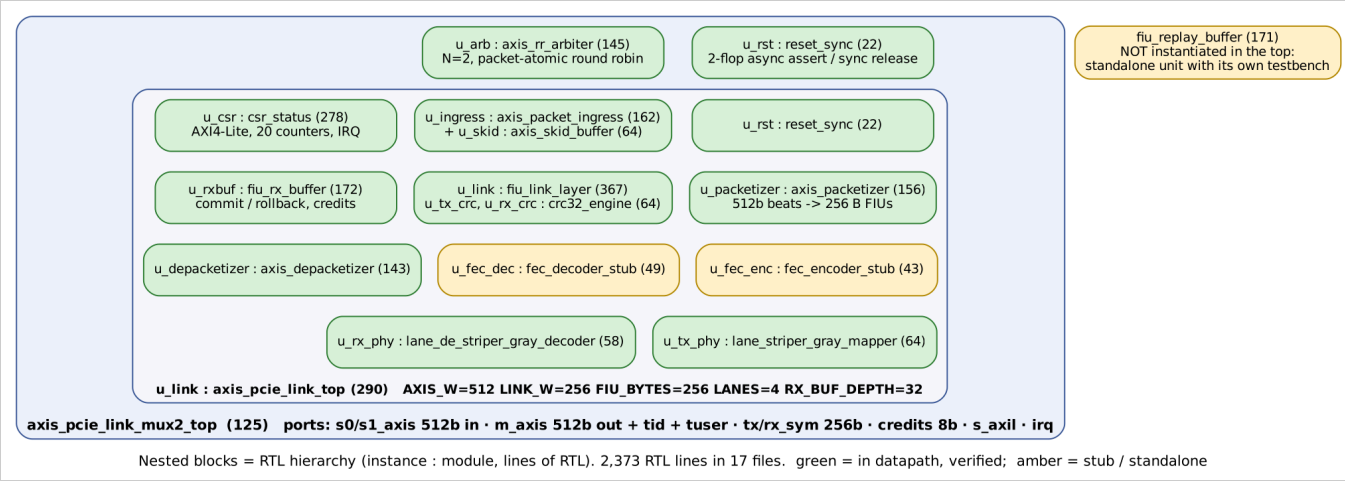


Fig. 2 — Module hierarchy (instance : module, RTL lines).

Interface (top)	Dir.	Width	Meaning
`clk`, `rst_n`	in	1, 1	Single clock; async assert / synchronised release
`s0_axis_*`, `s1_axis_*`	in	tdata 512, tkeep 64, tlast, tvalid / tready	AXI4-Stream slaves, one per source
`m_axis_*`	out	tdata 512, tkeep 64, tlast, tid, tuser	AXI4-Stream master; `tuser=1` = errored/truncated packet
`tx_sym` / `rx_sym` (+valid, sof)	out / in	256 + 2	4 lanes × 32 PAM4 symbols per clock (PHY-facing)
`rx/tx_credit_return`	out / in	8 / 8	Link-level credits (RX-buffer words)
`s_axil_*`, `irq`	slave / out	addr 8, data 32	AXI4-Lite CSR: control, 20 counters, W1C error status, IRQ

Total top-level port bits: **2,374** (1,482 in, 892 out) — this drives the die size (§6).

4. Verification

Requirement	Tests / checks	Result
Data transfer, end to end	sizes_directed, back_to_back, random_stalls, order_preservation, random_regression; VCD checks W1, W2, W8	**PASS, 8 seeds**
Register read/write (AXI4-Lite)	csr_access, err_csr_perf (shadow model of all 20 counters)	**PASS**
Reset / initialisation	reset_idle, reset_traffic, link_enable, arb_reset_traffic (reset with both sources mid-packet)	**PASS**
Error / corner conditions	tkeep_malformed, crc_corruption, symbol_errors, lane_corruption, header_forgery, arb_link_errors; W5–W7	**PASS**
Flow control	credit_stress (incl. link disable with credits exhausted), random sink back-pressure everywhere	**PASS**
Arbitration / fairness	8 arbiter tests; formal properties P1–P6 (k-induction); V1–V4	**PASS / PROVED**
Parameter robustness	LANES 1, 2, 8, 16, 32; TID_W 1, 8	**119/119**
Testbench strength	41 injected RTL bugs	**41/41 killed**

Netlist correctness	Formal mapping equivalence (pre/post ABC); gate-level simulation of synthesised and routed netlists (arbiter bench, 2 seeds)	**EQUIV.; 16/16 + 16/16**

Measured behaviour (RTL). 50.0 % share per source under saturation; small-packet source waits ≤ 159 cycles behind 4096-byte packets (bound 192); 7.93 cycles per back-to-back 1-beat packet (link bound 8); seed-1 error test: 600 packets in, 568 out, 95 CRC errors detected, every discard predicted by the reference model. Full table: `results/regression_summary.md` (338 rows, each with its log).

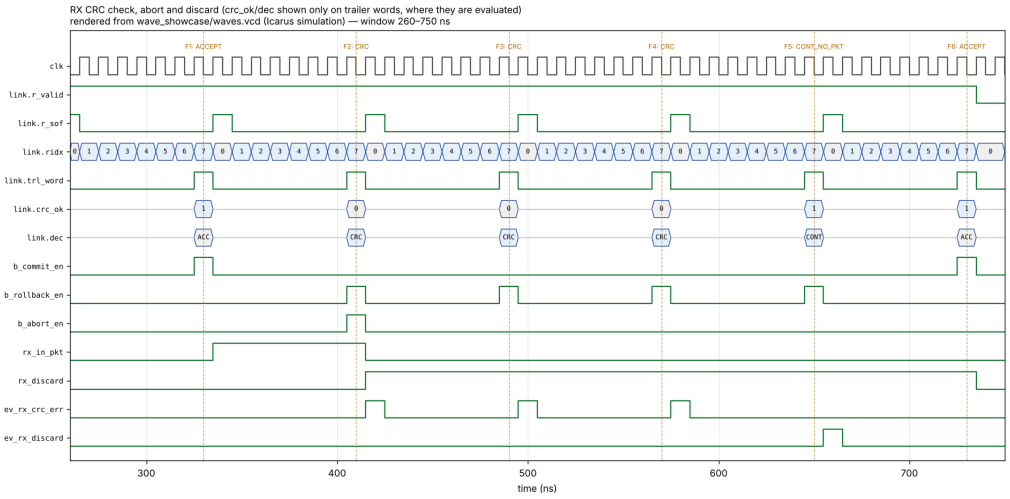


Fig. W6/W7 - receiver CRC check, abort and discard

crc_ok / dec are evaluated on trailer words: F1 ACCEPT (commit), F2-F4 CRC error (rollback; F2 also aborts the open packet), F5 continuation-without-packet (discarded), F6 ACCEPT. ev_rx_crc_err pulses once per bad FIU. Checks W6 (CRC recomputed from received words) and W7 (decision table vs DUT dec/commit/rollback/abort) - PASS.

Fig. 3 — Receiver CRC check, abort and discard after injected channel errors (more figures in `results/waveforms_annotated/`).

5. Synthesis (Yosys, Sky130 HD, TT 25 °C 1.8 V)

Metric	Value (source)
Standard-cell area	**594,554 µm²** (`synthesis/sky130_yosys/synth_stat.rpt`)
Cells / flip-flops	44,969 cells / 12,590 flip-flops
Sequential share of area	**62.08 %** — the 32 × 256-bit RX buffer and pipeline registers are built from flip-flops
Latches / unmapped cells / structural problems	0 / 0 / 0 (`check.rpt`)
Mapping equivalence (Yosys + ABC, register cut points)	**EQUIVALENT; negative control (one gate changed) NOT_EQUIVALENT; 12,590/12,590 flops matched**
Gate-level simulation of the synthesised netlist	**16/16 PASS**
Pre-layout STA	Not meaningful for sign-off: synthesis leaves high-fanout nets unbuffered (reset → 1,415 flops), so TT setup reads −32.0 ns; buffering in placement removes this (§6)

6. Physical design (OpenROAD, Sky130 HD)

Step	Settings and result (run `sky130hd_tt_100MHz`)
Floorplan	Die 1300 × 1300 µm, core margin 20 µm, 37.6 % utilisation from synthesis area. The die is **pin-limited**: 1,158 ingress pins on the north edge need ≥ 1,095 µm at 2-track pin pitch. Pins: met2 north/south, met3 east/west; PDN met1 rails + met4/met5 straps
Placement	Port buffers (`buf_2`) on all 2,366 I/Os; timing- and routability-driven global placement (density 0.55); repair with 20 % slew/cap margins
CTS	clkbuf_4/8/16 clock tree (1,797 `clkbuf` cells in the final design, tree + repair); hold repair with 0.05 ns margin
Global + detailed routing	met1–met5; detailed route converged 100,212 → 0 violations in 14 iterations (2 h 02 min); 3.88 m of wire
Antenna repair	392 violating nets → 0: 5 diode rounds (1,338 diodes) + targeted ECO (16 diodes placed beside the violating gate pins)
Extraction, IR drop, power	OpenRCX SPEF; static IR drop worst 0.23 mV on VDD (0.01 %); vectorless power 0.162 W at TT (default activity — not a sign-off figure)

Post-route STA (SPEF, propagated clock, 100 MHz)	Setup WNS	Hold WNS	Max slew / cap / fanout viol.	Result
TT 25 °C 1.80 V (optimisation corner)	+0.026 ns	+0.096 ns	0 / 0 / 20	**MET**
FF −40 °C 1.95 V	+2.116 ns	+0.055 ns	0 / 0 / 20	**MET**
SS −40 °C 1.40 V (stress corner)	−32.95 ns	−0.813 ns	23,910 / 0 / 20	**VIOLATED**

The SS library is the only slow corner that could be fetched (cold, 1.40 V — harsher than LibreLane's default `ss_100C_1v60`); the flow optimises at TT only. Gate-level simulation of the routed netlist (CTS buffers, resizing, hold buffers, diodes): **16/16 PASS**.

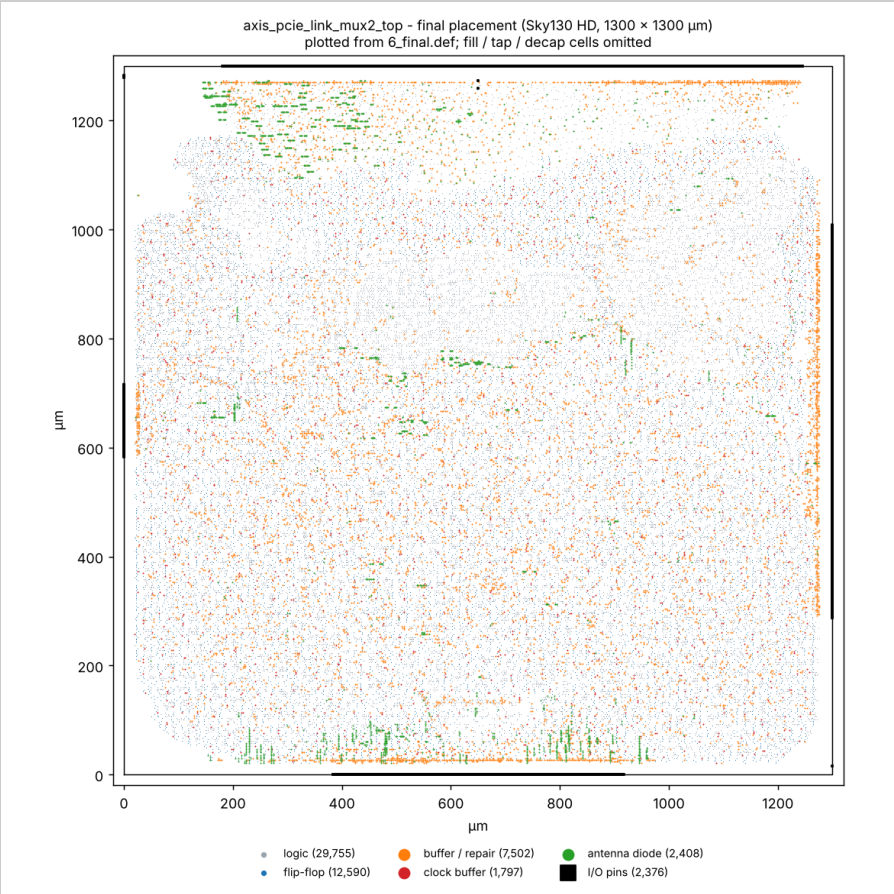


Fig. 4 — Final placement plotted from `6_final.def` (fill/tap/decap omitted). Antenna diodes cluster at the north/south pin edges; port and repair buffers line the edges.

7. Optimization summary — changes and measured impact

The first complete PnR run (attempt 1) met TT timing but left electrical-rule and antenna problems. Root causes were found and fixed in the flow, and the whole back end was re-run from placement (the recorded run). Both runs are in the build; numbers are from their reports.

Metric (post-route, TT)	Attempt 1	Final run	Change made
Max-slew violations	147	**0**	Repair with 20 % slew/cap margins at placement and after GRT, so diodes and extracted wiring added later still fit
Max-capacitance violations	19	**0**	
Max-fanout violations	26	20	`buffer_ports` on every I/O (input ports could not be repaired before)
Port buffer cell	—	`buf_2`	Explicit: left to choose, OpenROAD picked a clock-delay cell (`clkdlybuf4s50_1`)
Antenna nets after detailed route → final	295 → 0	392 → 0	Post-route ORFS-style diode loop + targeted diode ECO (new flow steps 5b, 5c)
Setup / hold WNS	+0.029 / +0.217 ns	+0.026 / +0.096 ns	Timing still met after the extra buffering
Standard-cell area (incl. buffers, diodes)	679,586 µm²	689,102 µm²	+1.4 %: cost of 2,366 port buffers and margins
Wirelength after detailed routing	3,909,262 µm	3,883,264 µm	−0.7 %
Vectorless power (TT)	0.162 W	0.162 W	unchanged

RTL-level changes carried into Stage 2 and identified next optimizations

- Stage-1 RTL hardening reused unchanged in Stage 2 (frozen with SHA-256 manifest): parameter legality checks that fail elaboration, single reset synchroniser, registered AXIS ingress (skid buffer), no latches.
- **Largest measured cost:** sequential cells are 62.08 % of synthesised area and the reset net fans out to 1,415 flops. Next RTL optimizations: RX buffer as an SRAM macro (OpenRAM/DFFRAM) instead of 32 × 256 flip-flops, and reset only on control state (not on datapath registers). Their effect has not been measured yet and is not claimed.

8. Sign-off status and open issues (honest)

Check	Result	Analysis / next step
KLayout DRC (ORFS sky130hd deck)	**1,522 violations**	1,520 × **m3.6** (met3 area < 0.24 µm²) + 2 × m3.1. Root cause verified in the routed database: the router stacks an `M2M3_PR` and an `M3M4_PR` via 0.23 µm apart, leaving only the merged via pads (≈ 0.19 µm²) on met3. The rule is in the tech LEF, but the router reported 0 violations — an independent check caught a router limitation. Fix: met3 patches at these stacks (or restrict stacked via pairs), incremental re-route, re-run DRC.
LVS (KLayout; Magic + Netgen)	not completed	Scripts complete and tested on single cells; the full-chip run (> 30 min on the 2-core host) was interrupted by a host restart and had not finished at submission.
GDS XOR, KLayout vs Magic	3,431 shapes	All differences are I/O pin rectangles at the die edge (counts match the pin counts per edge) and via-enclosure slivers on the met4/met5 power grid; no signal routing differs.
DRC engines themselves	**self-test PASS**	Both KLayout and Magic flag a planted 50 nm met1 error (negative control).

9. Limitations and roadmap

- **Close physical sign-off:** fix the met3 minimum-area stacks, finish LVS (both tool sets), clean the XOR pin-layer convention.
- **Multi-corner optimisation** and an SS sign-off library at a standard corner; reduce frequency or pipeline if SS must be met.
- **Area/power:** SRAM-based RX buffer; activity-based power (SAIF from simulation) instead of vectorless.
- **Architecture:** N×M crossbar, real FEC, integrated replay (the Go-Back-N buffer exists and is unit-tested), named network-packet and ML-tensor workloads with throughput reports.
- **Not attempted:** FPGA prototype (bonus), PCIe protocol layers and analog PHY.

10. Reproduction

RTL verification: `python3 sim/run_regression.py --seeds 1 2 3 4 5 6 7 8 --cov-strict`, `--bench arb`, `--bench replay`, `--sim verilator`, `python3 sim/run_mutation_check.py`, `bash sim/formal/run_formal.sh`. **ASIC flow:** `cd physical_design/flow && make platform pdk && make all` (tools: `tools/build_from_source.sh`, pinned commits). Every flow failure met and fixed is logged in `physical_design/flow_debug_log.md`. Exact commands and folder map: `README.md`.